

ReqRes API – Postman QA Assessment

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1. Overview

This Postman collection was created to test the ReqRes REST API and demonstrate API testing using different HTTP methods, authentication, environment variables, and automated test assertions.

The collection covers the complete user CRUD flow along with authentication and response validation.

2. Setup

A Postman environment named **ReqRes QA Environment** was created with reusable variables:

- `base_url` – Base URL of the ReqRes API
- `api_key` – API key used for API access
- `auth_token` – Authentication token returned from the login request
- `userid` – Stores the ID returned when a user is created

Using environment variables makes the collection reusable and avoids hard-coding values in individual requests.

3. Authentication

A login request is executed using the ReqRes authentication endpoint.

The test script validates the successful login response and extracts the returned token. The token is then stored dynamically in the `auth_token` environment variable.

Subsequent API requests use the token through the Authorization header:

Authorization: Bearer `{{auth_token}}`

This demonstrates token-based authentication and allows the authentication flow to be reused across the collection.

4. API Requests Covered

The collection includes:

- **POST – Login:** Authenticate and store the returned token
- **GET – List Users:** Retrieve and validate user data
- **POST – Create User:** Create a new user and validate the response
- **PUT – Update User:** Perform a complete update and validate the updated payload
- **PATCH – Update User Partially:** Modify selected user information and validate the response
- **DELETE – Delete User:** Delete a user and validate the expected response

5. Testing Approach

Automated Postman test scripts have been added to each request.

The tests validate:

- Expected HTTP status codes
- Response format and JSON structure
- Required response fields
- Request and response payload values
- Generated user IDs
- Created/updated timestamps where applicable
- Authentication token generation and storage
- Empty response body for DELETE where applicable

The collection is designed to support both individual request execution..

6. QA Approach

The testing focuses on both **functional API validation** and **response validation**. Instead of validating only the HTTP status code, the test scripts also verify important response data and business-relevant fields.

Environment variables are used to improve maintainability and reduce duplication, while automated assertions provide immediate feedback on whether each API request behaves as expected.

The collection and environment can be imported into Postman and executed using the configured environment.